SIGNALS AND DIGITAL DATA TRANSMISSION

- Data transmission or digital transmission is the physical transfer of data / information over a communication channel.
- In computer network, we send information form one computer to another.
- This information may be in the form of voice, image, characters or code , numeric data.
- In order to transmit this information across the network, it needs to be transformed into **electromagnetic signals**.
- Therefore in computer network information flows from one system to another in the form of **signals via transmission media.**
- This signal can be Analog or Digital form.

Signals

- A signal is
  - generated by a transmitter and transmitted over a medium
  - function of time
  - function of frequency, i.e., composed of components of different frequencies
- Analog signal
  - varies *smoothly* with time
  - E.g., speech
- Digital signal
  - maintains a constant level for some period of time, then changes to another level
  - E.g., binary 1s and 0s

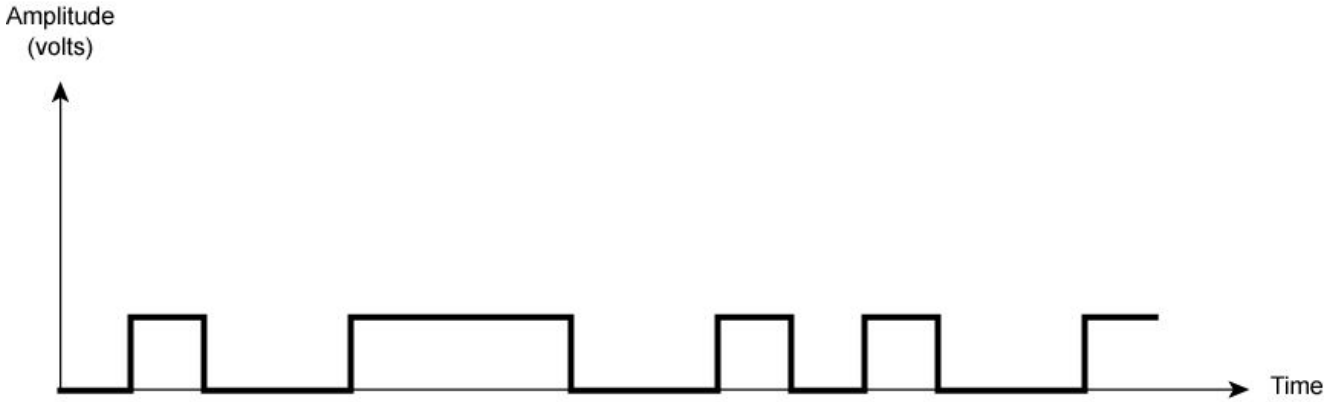
ANALOG TO DIGITAL

- Both data and the signals that represents them can either in Analog or Digital form.
- An Analog signal refers to continuous wave form that changes smoothly over time.
- Digital refers to discrete set of points of specific data.

- Signals can be analog or digital.
- Analog signals can have an infinite number of values in a range.
- Digital signals can have only a limited number of values.
- Analog signals have higher error rate.
- Digital signals have very less low error rate,
- Slow transmission when compared to digital signal
- Digital data has a faster rate of transmission.

Analog & Digital Signals

(a) Analog



(b) Digital

Periodic vs. Aperiodic Signals

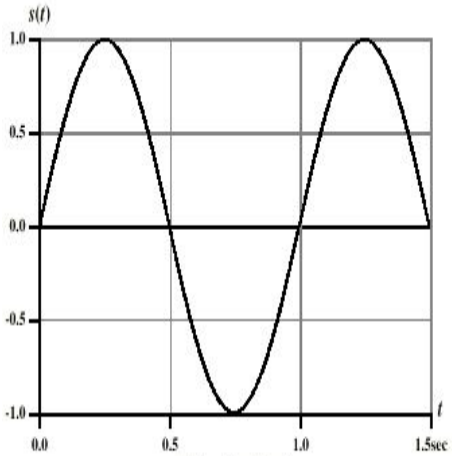
- A signal is periodic if it completes a pattern within a measurable time frame, called a period and repeats that pattern over identical period of time.
- The completion of one full pattern is called a cycle.
- A period is the amount of time required to complete one full cycle.

Aperiodic Signals

- An aperiodic or nonperiodic signals change constantly without exhibiting a pattern or cycle that repeats over time.

ANALOG SIGNALS

- Analog signals can be classified into simple or complex.
- A simple analog signal or a sine wave cannot be decomposed into simpler signals.
- A complex analog signal is composed of multiple sine waves.



(a) A = 1, f = 1, ϕ = 0

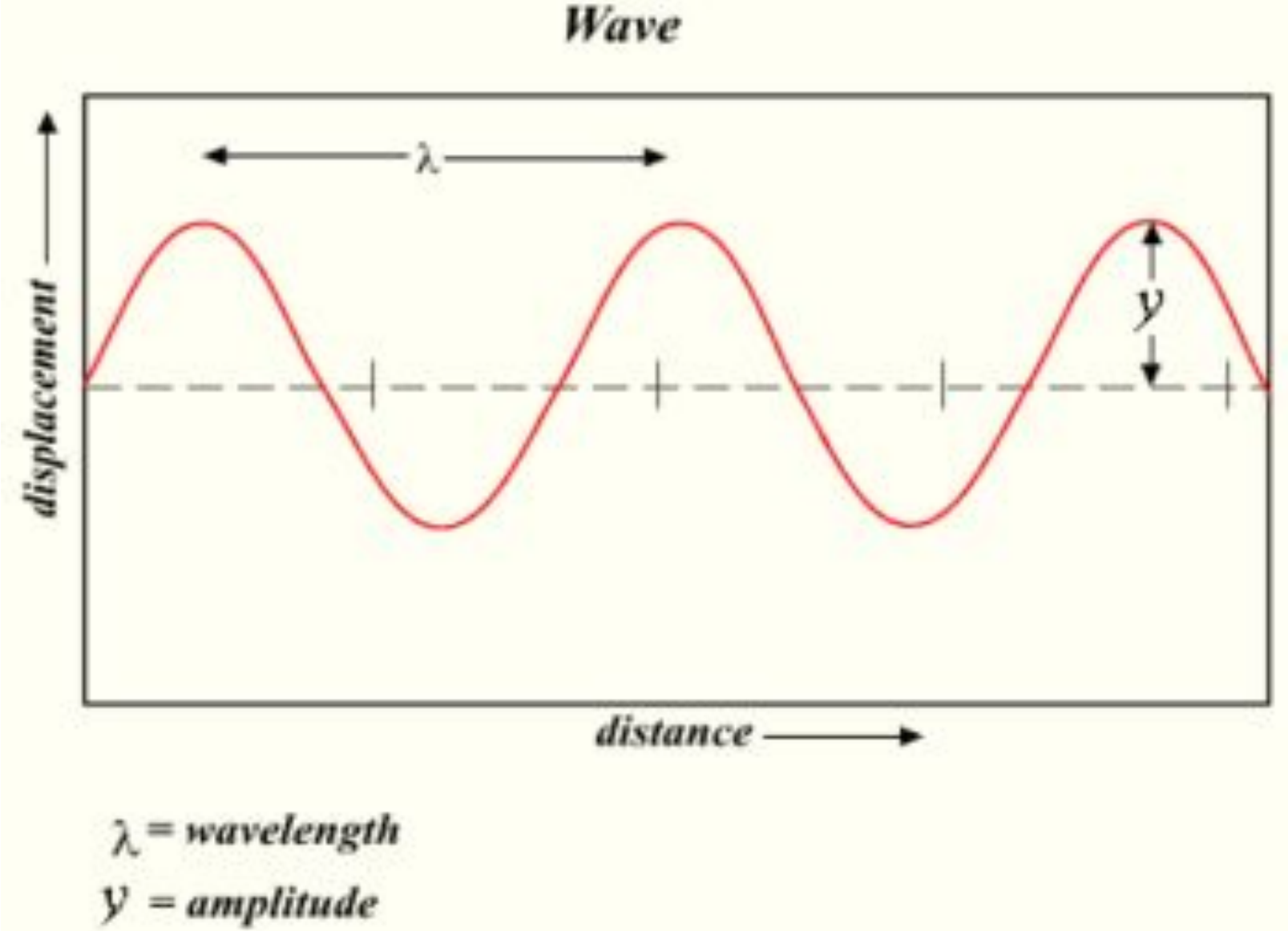
There are two types of wave forms are there :

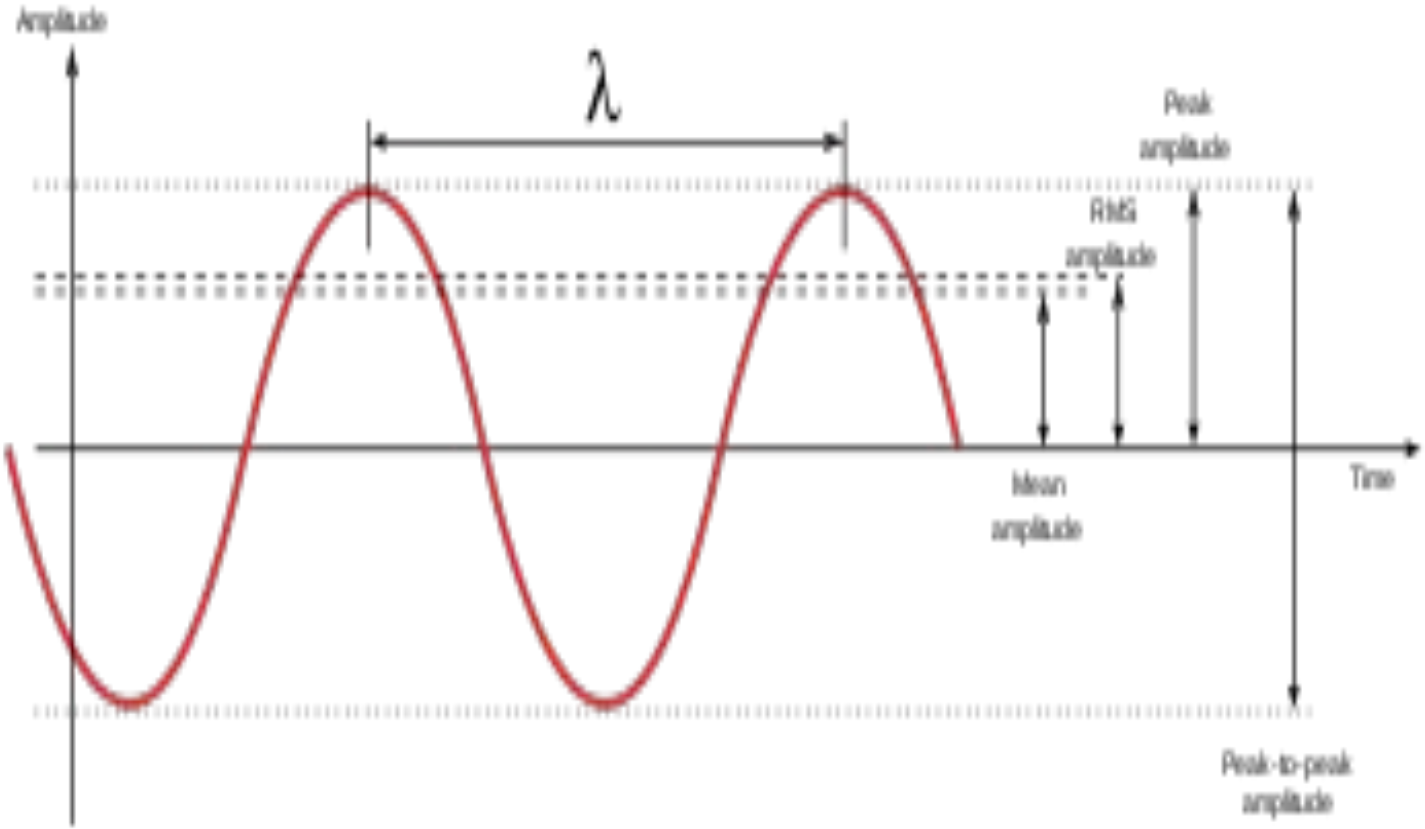
1. Sine Wave
2. Square Wave

Analog signals are described by 3 characteristic :- amplitude, frequency and phase.

1. **Amplitude :-** Amplitude of a signal refers to the height of the signal.

- It is equal to the vertical distance from a given point on the wave form to the horizontal axis.
- The maximum amplitude of a sine wave is equal to the highest value it reaches on the vertical axis.
- Amplitudes are always positive numbers (for example: 3.5, 1, 120).
- Amplitude is measured in either volts, amperes or watts depending on the type of signals.
- Volts is used for voltage, ampere for current and watts for power.


Wave
λ
y
displacement
distance
λ = wavelength
y = amplitude


Amplitude
λ
Peak
amplitude
RMS
amplitude
Mean
amplitude
Time
Peak-to-peak
amplitude

Frequency Period

- Period(T) is the amount of time a signal needs to complete one cycle.
- Frequency (f) is the number of cycles per second.
- **Frequency=1/period** **Period=1/frequency**
- It refers to the number of wave patterns completed in a given period of time.
- It refers to number of periods in one second or number of cycles per second.
- Frequency is measured in Hertz (Hz).
- Other units used to express frequency are kilo hertz, mega hertz and giga hertz and tera hertz.

Phase

- Phase(wave) describes the position of the wave form relative to time zero.
- Phase describes the amount by which the wave form shifts forward or backward along the time axis.
- Phase is measured in degree or radians.
- A phase shift of 360 degree indicates a shift of a complete period, a phase shift of 180 degree indicates a shift of half period and a phase shifts of 90 degree indicates a shift a quarter of a period.

Advantages of Analog Signals

- Best suited for the transmission of audio and video.
- To carry the information it consumes less bandwidth than digital signals

Digital Signal

- A digital signal is a discrete form.
- It can have only a limited number of defined values such as 1 and 0.
- Digital signals are represented by square wave.
- In digital signals 1 is represented by having a positive voltage and 0 is by no voltage or zero voltage.
- All the signals generated by computers and other digital devices are digital in nature.

Basic characteristics of digital signals :-

- Bit Interval :- It is the time taken to send one single bit. Instead of period it is used here.
- Bit rate:- It refers to the number of bit intervals per second usually expressed in bps(bits per second).

Advantages of digital signals

- Best used for the transmission of digital data.
- Digital data can be easily compressed.
- Digital information can be encrypted.
- Provides better clarity because all signals must be either 1's or 0's.

Digital representation of information

- The transfer of information of various types :-

1. some involves blocks of text characters like e-mail.
2. Others involve stream of information such as telephony.

1. Block-oriented information
2. Stream information

1. Block-oriented information

- Block oriented information occurs naturally in the form of a single block.
- These blocks of information may range from a few bytes to several hundred kilobytes and occasionally several megabytes.
- Normal files of this form usually contain fair amount of redundancies (like letter e is repeated several times in a text).
- Hence, data compression utilities such as compress, zip and other methods are used to encode the original information into blocks which will now take fewer bits to transfer and less disk storage space.

2. Stream Information

- Stream information is produced continuously and must be transmitted as it is produced.
- Theses signals are analog signals that are digitalized before transmission.
- Example :- Voice and music.

BASIC PROPERTIES OF DIGITAL TRANSMISSION SYSTEMS

- The purpose of a digital transmission system is to transfer a sequence of 0's and 1's from transmitter (on the left end) to a receiver (on the right).

DIGITAL TRANSMISSION SYSTEM

1. BIT RATE:- Is the transmission speed of bits measured in bits/seconds.

2. (Baud) Signal rate :- refers to the number of signal units per second that are required to represent those bits.

BAND WIDTH

- The band width of a channel is the range of frequencies that is passed by a channel.
- The band width is defined as the difference between the upper and the lower frequency limit of the signal or the equipment operation range.

f1 = Lower frequency – 300 Hz

f2 = Upper frequency – 3000Hz

Bandwidth = f2 – f1 = 3000-300 = 2700Hz.

- Bandwidth is expressed in terms of Hertz or in bits per second.

Signal – to – Noise Ratio

$$\text{SNR} = \frac{\text{average signal power}}{\text{average noise power}}$$

- SNR is the ratio of the signal power to the noise.
- Signal and noise both may change with time.
- High SNR – means the signal is less corrupted by noise.
- Low SNR – means the signal is more corrupted by noise.
- SNR is actually the ratio of what is wanted(signal) to what is not wanted (noise).
- Signal-to-noise ratio is defined as the powerSignal-to-noise ratio is defined as the power ratio between a signalSignal-to-noise ratio is defined as the power ratio between a signal (meaningful information) and the background noise (unwanted signal):

Fundamental limits in digital transmission

- The **quality of the digital transmission system** is determined by the transmission rate at which information bits can be transmitted reliably.

- It is measured in terms of 2 parameters :-

  1. Speed or bits/second

  2. Bit error rate.

1. Speed rate or data rate

- Data rate is the speed at which we can send data in bits per second.
- Data rate depends on 3 factors :-

1. **The bandwidth available**

2. **The level of signal we use.**

3. **The quality of the channel.**

Bit error rate

- Defines what percentage of bits are in error from the point where the data originates to the point where the user at the receiving end gets it.

- It can provide error rates of less than 0.0001 to 0.0002 percent.

- Two theoretical formulas were developed to calculate the data rate :-

1. Nyquist - for a noiseless channel

2. Shannon - for a noisy channel.

1. Nyquist - for a noiseless channel

- For a noiseless channel, Nyquist showed that the maximum signaling rate is twice the band width of the channel.

Bit rate = 2 * bandwidth * log2 L

where L is the number of signal levels used to represent data.

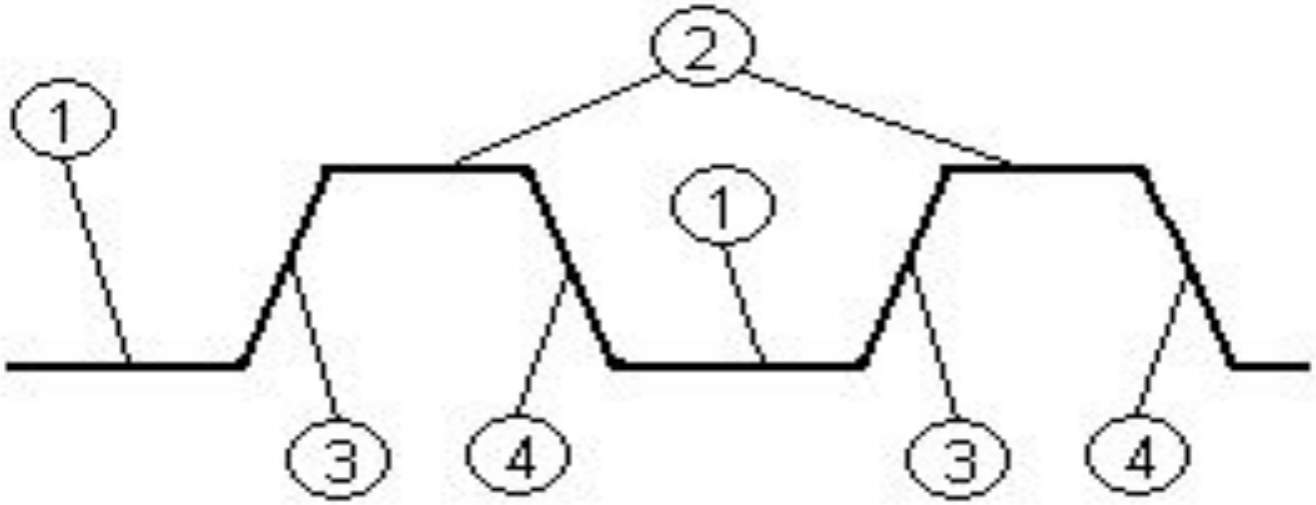
- Given a specific bandwidth, we can have any bit rate by increasing the number of signal levels.
- Although theoretically correct, practically there is a limit.
- When we increase the number of signal levels, we impose a burden on receiver.
- If the number of levels in a signal is 2, the receiver can easily distinguish between a 0 and 1.
- If the level of a signal is 64, the receiver must be very sophisticated to distinguish between 64 different levels.

2. Shannon - for a noisy channel

- The transmission channels are noisy.
- Two parameters that affects the performance of a digital transmission system are :- bandwidth and SNR.
- For a given channel of bandwidth and SNR the maximum channel capacity for a reliable communication is given by Shannon.

Capacity = bandwidth * log2 (1+SNR)

In the Shannon formula there is no indication of the signal level, which means that no matter how many levels are there.



A digital signal waveform: (1) low level, (2) high level, (3) rising edge, and (4) falling edge